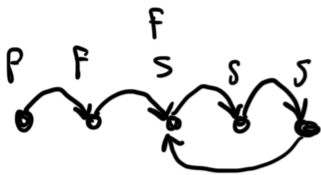


Find the beginning of a cycle in a list in $O(1)$ extra space.

Floyd's cycle finding:

two pointers $\rightarrow$ F 2 steps
 $\rightarrow$ S 1 step

found the cycle set F to init position and move it 1 step



! F move first

S $\rightarrow$ m steps
 F $\rightarrow$ 2m steps

$$h = a + b$$

before cycle inside cycle

$$2m = a + b + k \cdot \text{len}(\text{loop})$$

$$2a + 2b = a + b + k \cdot \text{len}(\text{loop})$$

$$a = k \cdot \text{len}(\text{loop}) - b$$

steps of F steps of S

Majority Element

$A[1, n]$

find, if any, the element that occurs at least $\frac{n}{2} + 1$ times

Boyer-Moore Algo:

A: 5 3 1 1 2 3 1 1 1

kill kill
 kill kill
 survive

- keep a candidate and a counter
- scan left to right
- $A[i] == c$ then counter++ else counter--
- counter == 0 $\Rightarrow c = A[i]; \text{counter} = 1$

if $\exists$ a majority element e then $c = e$ at the end

Why •

$\Theta(\log n)$ time and space

5 3 1 1 2 3 1 1 1

$\rightarrow$ 1 0 0 0 0 0 0 0

$\rightarrow$ 0 1 0 1 1 0 0 0

$\rightarrow$ 0 1 1 1 0 1 1 1

Majority

0
0
1

• most frequent

This support insertion and delete

Misra - Gries (Heavy Hitter)

$A[1, n]$

Find Elements that occurs $\frac{N}{T} + 1$ times

keep a set k with candidates with counters

for e in A

if e in k

$k[e] += 1$

else

$k[e] = 1$

if $\text{len}(k) > T$:

for k in $k.\text{keys}()$

$k[k] -= 1$

if $k[k] == 0$

del $k[k]$

Naïve analysis:

$O(T \cdot n)$ time $\rightarrow$ but actually is $\Theta(n)$ time

this cost
cannot be more
the number of
elements of the array